

National Coverage Determination (NCD)

Cardiac Output Monitoring by Electrical Bioimpedance

50-54

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Tracking Information

Publication Number

6

Manual Section Number

50-54

Manual Section Title

Cardiac Output Monitoring by Electrical Bioimpedance

Version Number

1

Effective Date of this Version

07/01/1999

Ending Effective Date of this Version

01/23/2004

Implementation Date

07/01/1999

Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

These devices utilize electrical bioimpedance to noninvasively produce hemodynamic measurements of cardiac output, specifically, stroke volume, contractility, systemic vascular resistance and thoracic fluid content.

Indications and Limitations of Coverage

Cardiac monitoring using electrical bioimpedance, a form of plethysmography, is covered, effective for services furnished on or after July 1, 1999, for the uses and conditions described below. Contractors should be aware that this technology is in the process of being proven for additional uses. Therefore, the uses below represent the current situation. Contractors may cover additional uses when they believe there is sufficient evidence of the medical effectiveness of such uses.

These devices are covered for the following uses:

1. Noninvasive diagnosis or monitoring of hemodynamics in patients with suspected or known cardiovascular disease;
2. Differentiation of cardiogenic from pulmonary causes of acute dyspnea;
3. Optimization of atrioventricular interval for patient with A/V sequential cardiac pacemakers;
4. Patients with need of determination for intravenous inotropic therapy;
5. Post heart transplant myocardial biopsy patients; and,
6. Patients with a need for fluid management.

Not covered at this time are the use of such devices for any monitoring of patients with proven or suspected disease involving severe regurgitation of the aorta, or for patients with minute ventilation (MV) sensor function pacemakers, since the device may adversely affect the functioning of that type of pacemaker. Also, these devices do not render accurate measurements in cardiac bypass patients while on a cardiopulmonary bypass machine, but do provide accurate measurements prior to and post bypass pump.

Covered uses of cardiac output monitoring by electrical bioimpedance should be billed using HCPCS code M0302.

Transmittal Information

Transmittal Number

109

Revision History

04/1999 - Limited coverage for six uses. Contractors may cover additional uses when they believe there is sufficient evidence of medical effectiveness of such uses. Effective date 07/01/1999. (TN 109) (CR 827)

08/1989 - Excluded from coverage. Effective date 08/25/1989. (TN 38)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001N\)](#)
- [First reconsideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001R\)](#)

Additional Information

Other Versions

Title	Version	Effective Between	
Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)	3	11/24/2006 - N/A	View
Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)	2	01/23/2004 - 11/24/2006	View
Cardiac Output Monitoring by Electrical Bioimpedance	1	07/01/1999 - 01/23/2004	You are here

